VISWANADHA ANURAG VISAGAKOTI

+91 9398939002 linkedin.com/in/anurag-viswanadha-visagakoti
anuragvvwork@gmail.com https://vvanurag.github.io/

SUMMARY:
AI systems engineer with strong ML and system thinking stemming from background in Robotics and Deep Learning. Brings hands-on experience specializing in stateful multi-agent orchestration, enterprise RAG architectures, and open-source model optimization. Proven track record deploying production-grade AI solutions including privacy compliant agent orchestration and LLM pipeline, voice agents on edge devices and parameter-efficient fine-tuning

SKILLS

- **Agentic AI & Orchestration**: LangGraph (StateGraph), LangChain (LCEL), Vercel AI SDK (ToolLoopAgent), Multi-Agent Architectures, Tool Calling & Structured Outputs, MCP (Model Context Protocol), Human-in-the-Loop (HITL).
- **RAG & Information Retrieval**: Retrieval-Augmented Generation (RAG), ChromaDB, Semantic Chunking, Metadata-Level RBAC Filtering, Context Window Optimization, Token Pruning.
- **Model Fine-Tuning & Quantization**: Parameter-Efficient Fine-Tuning (LoRA, QLoRA), SFT (TRL SFTTrainer), Quantization (bitsandbytes, AWQ, GGUF), 4-bit/8-bit, Hugging Face (Transformers, Accelerate), PyTorch
- **Inference Serving & Optimization**: vLLM (PagedAttention), Ollama, FlashAttention-2, TensorRT, ONNX Runtime, CUDA Kernel Optimization, Dynamic Batching.
- **Computer Vision & Edge AI**: 3D Gaussian Splatting, NeRF, TFLite Micro (Embedded Wake-Word Models), ESP32, OpenCV, Point Clouds, SLAM, ROS/ROS2
- **Robotics, Navigation & Control**: ROS / ROS2, Motion Planning (A*, OMPL), Multi-Body Dynamics, Kinematics, Visual Servoing, Sensor Fusion (EKF, Factor Graphs), Dual-Arm Manipulators, Quadrotors, Mobile Robots (UGV)
- **Robotics Simulation & Sensors**: NVIDIA Isaac Sim, Gazebo, CARLA, RViz, MATLAB / Simulink; LiDAR, RGB-D / Stereo Cameras, IMU, RTK-GNSS, Topcon.

- **Languages & Production Systems**: Python, C++, SQL, PowerShell, FastAPI, Pydantic v2, Tenacity (Resilience), Docker, AWS (EC2 G4/G5), Linux, Git

INDUSTRY EXPERIENCE

AI enabled Project coordinator: NirvanaHealth | RxAdvance, Southborough, USA Oct 2025 – Current
Agentic AI automation layer over Azure Boards

- Coordinated project delivery for a **50-member technology team** supporting **health insurance platforms**.
- Introduced **productivity and execution-tracking metrics**, improving visibility into project progress and enabling teams to track measurable outcomes.
- Implemented an **Agentic AI automation layer on top of Azure Boards**, streamlining project tracking, status reporting, and operational workflows.
- Architected an air-gapped, **zero-egress DevOps intelligence pipeline** using local open-source LLMs (**Llama 3** via **Ollama**) and **LangChain**, ensuring 100% data privacy and strict HIPAA/PHI **compliance** with zero cloud data exposure.
- Engineered a **multi-agent system using LangGraph (StateGraph)** coordinating specialized retrieval, context synthesis, validation, and correction agents; implemented dynamic error-recovery loops that autonomously healed malformed JSON outputs, achieving 99.9% schema compliance against strict **Pydantic** v2 contracts.
- Developed a high-throughput TFS ingestion engine translating WIQL queries into batch REST calls with **payload pruning**, **slashing LLM prompt token consumption** by 82% and eliminating redundant API traffic.
- Constructed an enterprise architecture knowledge retrieval (**RAG**) pipeline using **ChromaDB** and **semantic chunking** (RecursiveCharacterTextSplitter), enforcing database-level Role-Based Access Control (**RBAC**) via metadata filtering.
- Integrated **Tenacity retry policies** with exponential backoff and randomized jitter to mitigate transient 5xx network spikes and rate limits across internal APIs, reducing pipeline execution failures to 0%.

Software Engineer: Technofix IT Solutions, Tempe (Remote), USA July 2025 – Sept 2025
Voice assistant agentic AI

- Engineered an **autonomous conversational agent loop (Voice assistant)** using **Vercel AI SDK**, executing **dynamic multi-step tool calls** for timers, reminders, and search.
- Implemented a **streaming agent orchestration layer** with dynamic tool execution and boundary clipping to optimize text-to-speech synthesis.

- Built an end-to-end voice pipeline integrating **Whisper/Fish-Audio STT, LLM reasoning, and OpenAI TTS** with real-time PCM16 audio resampling.
- Formulated a deterministic **5-stage session state machine** coupled with **token-bucket rate limiting** and **privacy-preserving** client IP hashing for audit logs.
- Architected a low-latency voice streaming gateway using **WebSockets** and a custom 20-byte binary protocol (OLX2).
- Integrated edge-to-cloud voice streaming on ESP32 microcontrollers, deploying TFLite micro wake-word models.
- Designed an SQLite persistence layer supporting forward schema migrations, multi-version assistant reply branching and regeneration (addReplyVersion), and automated context window budgeting that pruned heavy base64 image attachments while preserving dialogue history.

Monitoring Analyst: Public Partnerships LLC, Tempe (Remote), USA July 2024 – July 2025

Payroll automation

- Engineered **SQL-based automation pipelines** for end-to-end auditing of timesheets, fund allocation, and payment disbursement, reducing manual intervention by over 60%.
- Designed **Python scripts** for parsing NAV audit logs, generating real-time status reports, and identifying high-priority exceptions using regex and structured log aggregation.

AI/Robotics Research assistant: Arizona State University, Tempe, USA June 2023 – July2024

Path planning optimization in Gaussian splatting 3D environment

- **Fine-tuned** Llama 2 (7B) and Mistral using **QLoRA** (4-bit NF4) on 30K+ spatial instruction pairs to translate natural language goals into 3D navigation waypoints.
- Quantization & Efficiency: Applied 4-bit AWQ/GGUF quantization with **bitsandbytes**, cutting VRAM requirements from 16GB to 5.5GB for single-GPU deployment.
- 3D **Gaussian Splatting** Integration: Coupled the fine-tuned spatial planner with 3D Gaussian Splatting scene models, optimizing CUDA rasterization for real-time obstacle avoidance.
- High-Throughput Serving: Served the quantized spatial LLM using vLLM (PagedAttention) and FlashAttention-2, boosting token throughput by 2.8x to meet robotic control loop constraints.
- Curated synthetic multi-view spatial navigation datasets in PyTorch, evaluating plan fidelity against baseline A* and trajectory benchmarks.

Staff Engineer: Treadstone Media Labs, Bengaluru, India March 2022 – May 2022

- Collaborated to **define requirements and plan projects, optimizing delivery** under **limited resource constraints**.
- Led **end-to-end planning and execution** of **interactive graphics tooling**, coordinating research, development, and deployment milestones for a production-grade animation pipeline used by creative and procedural content teams.
- **Optimized** the PiFUHD deep implicit surface network by converting **PyTorch models to ONNX** and compiling TensorRT FP16 inference engines, cutting per-image 3D reconstruction latency.
- Architected and **deployed a containerized GPU inference** service on AWS EC2 (G4dn/G5 instances) using **Docker** and **FastAPI** with asynchronous worker queues.
- Planned and delivered **deep learning–based 3D asset generation tools** using **PiFUHD**.
- Defined **data strategy and execution plan** for a proprietary 3D dataset (>100K meshes), coordinating **dataset curation, augmentation, and model fine-tuning** (**transfer learning, LoRA**) to meet quality and generalization targets.
- Designed and coordinated **rollout of a scalable AWS-based inference system (EC2 G4/G5)**.
- Planned and implemented containerized delivery of the tooling suite using Kasm Workspaces, providing browser-based, GPU-backed environments for distributed artists and technical directors.

Lead Engineer: Orange Bodhi Technologies, Hyderabad, India Apr 2020 – March 2022

- Architected and deployed a fully automated **multi-stage chemical etching system** using a **4-DOF robotic manipulator** to transport metal substrates through sequential acid bath treatments, each requiring dip time and agitation profiles.
- Designed and implemented a **PLC logic stack** with **real-time HMI interfacing** for system operators to monitor acid concentration, tank temperatures, and stage timings, via **ladder logic** and structured text.
- Developed a **state-machine** based process controller that synchronized robotic arm motions with tank-specific safety interlocks and dynamic timeout triggers based on system sensor feedback.
- Engineered **stereo vision**-based surface inspection modules using calibrated industrial cameras and custom OpenCV pipelines to detect surface inconsistencies (bubbles, discoloration, erosion), enabling closed-loop feedback to optimize dip durations and motion paths.
- Implemented **dynamic robot path planning** routines using **inverse kinematics in ROS** to adapt to conveyor line offsets, ensuring consistent submersion across variable workpiece geometries.

System Engineer: TCS Innovation Labs, Kolkata, India March 2017 – April 2020

Retail Space Automation with quadrotors

- Conducted R&D of **3D perception** and **navigation** for quadrotor-based retail space automation.
- Designed and constructed a **high-fidelity indoor testing facility** equipped with a **Vicon motion capture system** for ground-truth trajectory recording, enabling quantitative evaluation of localization accuracy, loop closure, and drift correction in real-time.
- Tested SOTA **path planning(A*)** and **trajectory control** to operate indoor quadrotors using motion capture feedback.
- Modified **ORB-SLAM** code for higher feature output for line-based object segmentation pipelines (**Line 3D++**).
- Integrated **DfUSMC (Depth-filtered Unsupervised Surface Mesh Completion)** into the navigation stack, enabling detailed 3D reconstruction during hover states and fly-throughs, supporting semantic segmentation.
- Developed a telemetry bridge between **PX4 autopilot firmware** and the perception stack using **MAVROS**, allowing direct interfacing of high-level motion commands from SLAM to the flight controller.

Robotics Engineer: Asimov Robotics, Kochi, India Feb 2016 – Feb 2017

Design and development of PR humanoid robot with custom manipulators

- Led the **end-to-end mechanical design and control integration** of a humanoid robot targeted for public interaction roles in malls, hospitals, and expos, focusing on aesthetic ergonomics and dynamic functionality.
- Implemented **manipulator control** for basic tasks considering payload by formulating manipulator dynamics.
- Executed **2D LiDAR SLAM** based navigation based on pre-built map to provide customer assistance.

Research Assistant: IIIT Hyderabad, Hyderabad, India Feb 2012 – Jan 2016

Floating base Dual arm space manipulator-Kino dynamic Simulation, planning and control

- Research involved **Task priority-based manipulation**, **multi-body dynamics** formulation, **visual servo control**, **Planning** in image space for a floating base redundant dual-arm space manipulator. ICRA[3], IROS[1], ICAR[2].
- Innovated new methods in calculation of **dynamic** decoupling of floating base disturbance and dual arm manipulator.
- Incorporated in MATLAB simulation using numerical methods and employed **visual servo control**
- Employed image space path planning for optimized execution while minimizing base disturbance.
- Constructed a prototype to demonstrate gravity free conditions on a hyper-smooth granite slab using air pressure knobs.

Product Design Engineer: Purnansh Design, Kochi, India Jun 2010 –Dec 2011

Product Design: Aesthetic, functional and CAD Design

- Designed consumer-grade and industrial products with a focus on manufacturability, integrating both aesthetic form and functional mechanical performance.
- Created detailed 3D CAD models and production-ready drawings in **SolidWorks**, applying GD&T and tolerance stack-up analysis to ensure part interchangeability.
- Collaborated with fabrication vendors to translate digital models into prototypes using injection molding, sheet metal, and 3D printing workflows.
- Conducted design validation via **FEA simulations**, ergonomic analysis, and DFM/DFA reviews to balance structural integrity with visual appeal.

EDUCATION

- **M.S. Robotics and Autonomous Systems (Systems Engineering)** May 2024
- Arizona State University, Tempe, AZ
- **M.S. Computer Science and Engineering** July 2015
- International Institute of Information Technology (IIIT), Hyderabad, India
- **B.Tech. Mechanical Engineering** June 2010
- Vellore Institute of Technology (VIT), Vellore, India

PUBLICATIONS

- [1] Rachit Bhargava, P.Mithun, V. V. Anurag, A.H.Abdul Hafez, S. V. Shah "Image Space based Path Planning for Reactionless Manipulation of Redundant SpaceRobot" IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS),Korea, October 2016.
- [2].Shah, S. V., Viswanadha Visagakoti, Anurag, Abdul Hafez, A. H., "Switching method to Avoid Algorithmic Singularity in Vision-based Control of a Space Robot", International Conference on Advanced Robotics(ICAR), IEEE, May 2015, Istanbul, Turkey
- [3] Abdul Hafez, A. H., Anurag, V. V., Shah, S. V., Krishna, K. M., Jawahar, C. V."Reactionless visual servoing of a dual-arm space robot", International Conference on Robotics and Automation(ICRA), IEEE,May 2014, Hong Kong, China